

Vivek Kumar Mishra

Curriculum Vitae — EB-2 National Interest Waiver Petition

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PROFESSIONAL SUMMARY

AI/ML engineer and IEEE Senior Member specializing in resilient distributed systems for U.S. critical infrastructure. Currently architects high-availability platforms at JPMorgan Chase & Co. (Systemically Important Financial Institution) serving **80+ million** American consumers with **99.9%** reliability. Author of 9 peer-reviewed publications (H-Index: 4, 37+ citations) with two permanently archived in the U.S. National Transportation Library. Technical book author with academic institutional adoption. Peer reviewer for NeurIPS 2025 and ICLR 2025.

EDUCATION

Master of Science in Computer Science **August 2021**
California State University, Long Beach — R2 Doctoral University (High Research Activity) *GPA: 3.8/4.0*

Bachelor of Technology in Information Technology **May 2014**
West Bengal University of Technology (MAKAUT) — Kolkata, India

PROFESSIONAL EXPERIENCE

Associate Software Engineer **October 2021 – Present**
JPMorgan Chase & Co. — Wilmington, Delaware *SIFI: \$4.4T assets, 84M consumers*

- Founding engineer on critical financial infrastructure platform serving **80+ million** Chase card-members
- Designed high-availability APIs (Java, Spring Boot, AWS) processing **5M+ requests/month** at 99.9% reliability
- Architected fraud prevention service integration protecting consumer accounts at enterprise scale
- Built observability and alerting infrastructure for mission-critical SIFI systems
- Improved application performance by **43%**; increased test coverage from 17% to **97%**

Research Assistant **March 2020 – August 2021**
California State University, Long Beach *Recruited for AI/Python expertise*

- Led U.S. DOT-funded intelligent transportation research (Mineta Transportation Institute)
- Two publications permanently archived in the National Transportation Library (ROSA P dot/61210, dot/59030)
- ML methodologies adopted by Florida DOT in federally funded infrastructure programs
- Research supports **Justice40 Initiative** goals for transportation equity

Member of Technical Staff **July 2018 – January 2020**
Oracle Financial Services Software — Bangalore, India

- Automated Oracle Cloud Infrastructure testing with Ansible, Python, and Go

- Infrastructure-as-code solutions using Terraform and Docker for distributed cloud systems

Programmer Analyst

August 2014 – July 2018

Cognizant Technology Solutions — Bangalore, India

- Automated 500+ test cases, reducing CI runtime by 44%
- Global Project of the Year Award

SCHOLARLY PUBLICATIONS (9 PEER-REVIEWED • 37+ CITATIONS • H-INDEX: 4)

1. AI-Driven Optimization of Backend and Cloud Infrastructure in Large-Scale Financial Systems. *IJARSC T* (IF: 8.2), Feb 2026.
2. Integrating Solar Photovoltaic Systems into the Grid: An Overview of AI Application.
3. The Role of Generative AI in Revolutionizing Healthcare, Education, and Finance.
4. MDPI Sustainability (IF: 3.9), sustainability-16-08077.
5. Framework for Rail Transport Inequality Assessment — **15 citations**.
6. Evaluating Financing Mechanisms and Economic Benefits to Fund Grade Separation Projects — **12 citations**.
7. Evaluating Innovative Financing Mechanisms for the California HSR Project — 7 citations. *NTL indexed*.
8. Funding the High-Speed Rail: A Case Study of the California Project — 3 citations.
9. Optimizing Multimodal Transportation Access for Low-Income Transit Riders. *NTL indexed*.

TECHNICAL BOOK

Design Patterns for AI Agents — ISBN: 979-8268912371

A Structured Approach to Building Robust and Intelligent Systems

Adopted by Anand International College of Engineering (AICTE-approved) for curriculum use.

PROFESSIONAL RECOGNITION

- **IEEE Senior Member** (#100495418, Nov 2024) — Top 8% of 460,000+ members
- **AWS Certified Cloud Practitioner**
- **IEEE Senior Member Review Panel** — Evaluating membership applications (June 2025)
- **IEEE Transmitter Impact Creator / IEEE.tv Speaker** — Official thought leadership platforms

PEER REVIEW SERVICE

- **NeurIPS 2025** — Position Paper Track Reviewer (H5-index: 278)
- **ICLR 2025** — FM-Wild Workshop Reviewer (H5-index: 362)

- **IEEE Senior Member Application Review Panel** — June 2025

NATIONAL INTEREST ALIGNMENT

- **Executive Order 14110:** Production AI systems at SIFI, AI agent frameworks, and peer-reviewed research advance safe, trustworthy AI deployment.
- **SIFI Protection:** Infrastructure protecting 80M+ consumers at institution critical to U.S. financial stability (Dodd-Frank / FSOC).
- **Justice40 Initiative:** Transportation research delivering infrastructure benefits to disadvantaged communities.
- **Federal Knowledge Infrastructure:** Two publications in National Transportation Library supporting federal investments.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, Go, Shell
Frameworks: Spring Boot, Spring Security, React, TensorFlow, Pandas
Cloud/Infra: AWS (DynamoDB, SQS, ECS, S3, IAM), Terraform, Docker, K8s
AI/ML: LLMs, GenAI, NLP, Agent-Based Modeling, OpenCV
Security: OAuth 2.0, OIDC, SAML, JWT
Monitoring: Splunk Enterprise, Observability, Alerting Systems

HONORS & AWARDS

- Eminent Speaker — AICTE-approved Faculty Development Programme
- Global Recognition Award 2024
- #FutureOfMeterReading Contest Winner — Capgemini Super Techies
- Global Project of the Year Award — Cognizant Technology Solutions